

Audio Processing in NLP

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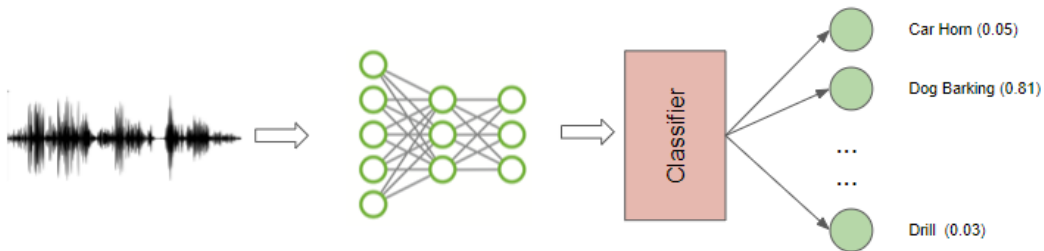


Image source: https://miro.medium.com/v2/resize:fit:1400/1*3MadXYXhtHr-XZNP5-1GXg.png

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Speech is the **most natural form of communication**.

What: Where speech interfaces are essential.

- ▶ Voice assistants (e.g., Amazon Echo, Siri, Google Assistant)
- ▶ Accessibility: captions, screen readers
- ▶ Hands-free operation (cars, IoT)
- ▶ Multilingual interaction & translation

Why: Speech enables more intuitive and inclusive interaction.

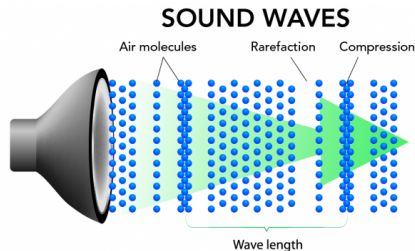
Why audio needs its own toolkit.

- ▶ Audio data is abundant and diverse, encompassing speech, music, and environmental sounds.
- ▶ Text-oriented methods do not apply directly: audio arrives as a continuous waveform with no tokens, no word boundaries, and thousands of samples per second.
- ▶ Many NLP systems now start from speech, not just text (e.g., live captions, real-time translation).
- ▶ The signal carries information text discards: speaker identity, prosody, emotion, and overlapping speakers.

By the end of this session, you will be able to:

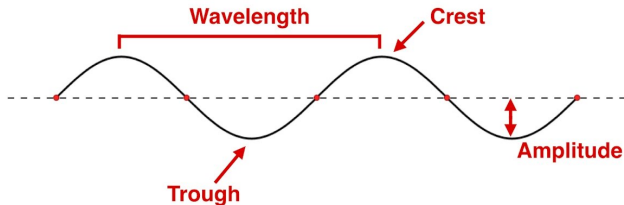
- ▶ Explain how a sound wave becomes a digital signal (sampling, quantization, bitrate).
- ▶ Explain how raw audio becomes features (MFCC, Mel/Log-Mel).
- ▶ Read & reason about spectrograms (STFT, mel scale).
- ▶ Compare the main ASR architectures: CTC, attention-based encoder–decoder, transducers.
- ▶ Describe the Text-to-Speech (TTS) pipeline and where its difficulty lies.
- ▶ Connect features $\rightarrow$ ASR/NLU/TTS.
- ▶ Evaluate a system with the right metric (WER, CER, DER, MOS).
- ▶ Recognize the challenges and future directions in audio processing for NLP.

Fundamentals of Sound



Source: Let's Talk Science, "What is Sound and How do we Hear it?"

- ▶ Sound is a type of energy produced by vibrating objects.
- ▶ It travels through air (or other media) as waves.
- ▶ Sound waves are longitudinal waves, meaning the particle displacement is parallel to the direction of wave propagation.



Source: "Sound and Audio," COMD1112, OpenLab at City Tech, CUNY

- ▶ **Waveform** $x(t)$: The shape of the sound wave, representing how air pressure changes over time.
- ▶ **Frequency**: Number of cycles (oscillations) per second, measured in Hertz (Hz). Determines the pitch of the sound.
- ▶ **Amplitude**: The height of the wave, representing the loudness or intensity of the sound.

- **Pitch/Formants:** The perceived frequency of a sound and its resonant frequencies, respectively.

To convert an analog signal into a digital one, two main steps are used:

- ▶ **Sampling:** The process of taking measurements of the continuous signal at regular time intervals.
- ▶ **Quantization:** The process of rounding off the amplitude of each sample to the nearest value on a finite scale.

The result is a **discrete signal**, denoted as $x[n]$, which is a sequence of numbers representing the original analog signal.

Why do we digitize signals?

- ▶ **Computer Processing:** Computers can only understand and manipulate discrete numerical values.
- ▶ **Reproducibility:** Digital signals can be copied and transmitted without any loss of quality.
- ▶ **Processability:** Digital signals are easier to process and manipulate using various algorithms for tasks such as noise reduction, filtering, and compression.

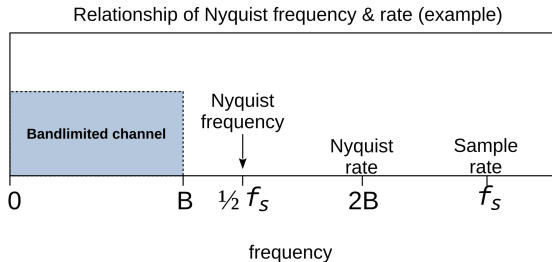
- ▶ **Sampling Rate** (f_s): The number of samples taken per second, measured in Hertz (Hz) or kilohertz (kHz). Common values include 16 kHz for speech and 44.1 kHz for music.
- ▶ **Nyquist–Shannon Sampling Theorem**: States that to perfectly reconstruct a continuous signal, the sampling rate (f_s) must be at least twice the maximum frequency ($f_{\max}$) of the signal.

$$f_s \geq 2f_{\max}$$

Failing to meet this condition can cause **aliasing**, where high frequencies are incorrectly represented as lower frequencies. The Nyquist limit, also known as the Nyquist frequency, is defined as half of the sampling rate:

$$f_{\text{Nyq}} = \frac{f_s}{2}$$

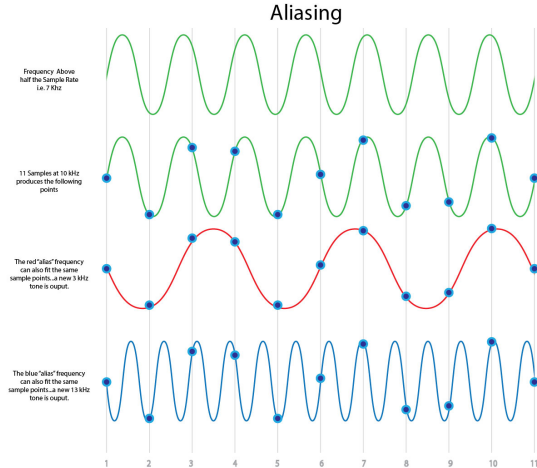
- ▶ **Quantization**: The process of mapping continuous amplitude values to a finite set of discrete levels. This process introduces **quantization noise**, which is a form of error.



Source: Wikipedia, "Nyquist rate".

- ▶ **Aliasing:** If the sampling rate is too low, higher frequencies are misrepresented as lower frequencies, causing distortion.

Digitizing speech: sampling, quantization, bitrate (cont.)



Source: M. Waldrep (Dr. AIX), "Aliasing and Foldover: A Primer," Real HD-Audio, 2014.

- ▶ **Bit Depth:** Common depths include 8-bit (256 levels), 16-bit (65,536 levels), and 24-bit (16.7 million levels). Higher bit depth allows for more precise representation of amplitude.
- ▶ **Bitrate:** The amount of data processed per second, calculated as:

$$\text{Bitrate} = \text{Sampling Rate} \times \text{Bit Depth} \times \text{Number of Channels}$$

For example, a stereo audio file with a sampling rate of 44.1kHz and a bit depth of 16-bit has a bitrate of:

$$44,100 \times 16 \times 2 = 1,411,200 \text{ bits per second} = 1.4112 \text{ Mbps}$$

Pre-processing Pipeline

1. Pre-emphasis Filter

- ▶ Enhances high-frequency components of the audio signal.
- ▶ Equation:

$$y[n] = x[n] - \alpha x[n - 1]$$

where $x[n]$ is the input signal, $y[n]$ is the output, and α is typically between 0.95 and 0.97.

- ▶ Helps balance the frequency spectrum and improves feature extraction.

2. Framing

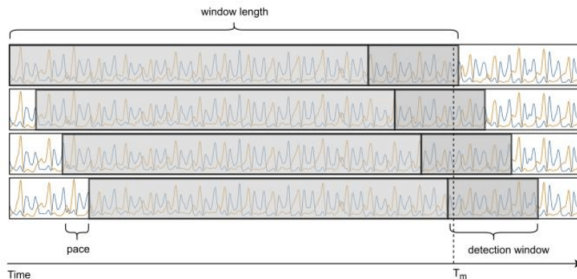
- ▶ Audio signals are divided into short frames (e.g., 25 ms) to capture local temporal features.
- ▶ Each frame overlaps with the next (commonly a 10 ms hop, i.e. 15 ms overlap).

3. Windowing

- ▶ Each frame is multiplied by a window function (commonly Hamming window) to reduce spectral leakage.
- ▶ Hamming window equation:

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$$

where N is the frame length.



The sliding-window scheme in general form: a fixed window length advanced by a fixed step. Framing an audio signal is the same construction, with a 25 ms window and a 10 ms hop.

Source: C. Morbidoni, A. Cucchiarelli, V. Agostini, M. Knaflitz, S. Fioretti and F. Di Nardo, "Machine-Learning-Based Prediction of Gait Events From EMG in Cerebral Palsy Children," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 819–830, 2021, Fig. 2 (CC BY 4.0). The traces are EMG, not speech.

Summary of Pre-processing Steps

1. **Pre-emphasis:** Boosts high frequencies.
2. **Framing:** Splits signal into short, overlapping segments.
3. **Windowing:** Applies Hamming window to each frame.

STFT & Spectrograms

- **Discrete Fourier Transform (DFT):** A mathematical operation that converts a signal from the time domain to the frequency domain. It transforms N samples of a discrete signal, $x[n]$, into N frequency bins, $X[k]$. The formula for the DFT is:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

where $x[n]$ is the signal in the time domain, $X[k]$ is the signal in the frequency domain, N is the total number of samples, n is the sample index, and k is the frequency bin index.

- **Fast Fourier Transform (FFT):** This is not a different transform, but rather a computationally efficient algorithm for calculating the DFT.

To apply the Fourier transform in practice, you typically:

1. **Choose a window length (N):** This determines the number of samples to be transformed at a time.
2. **Choose a hop size:** This is the number of samples to advance before taking the next window.

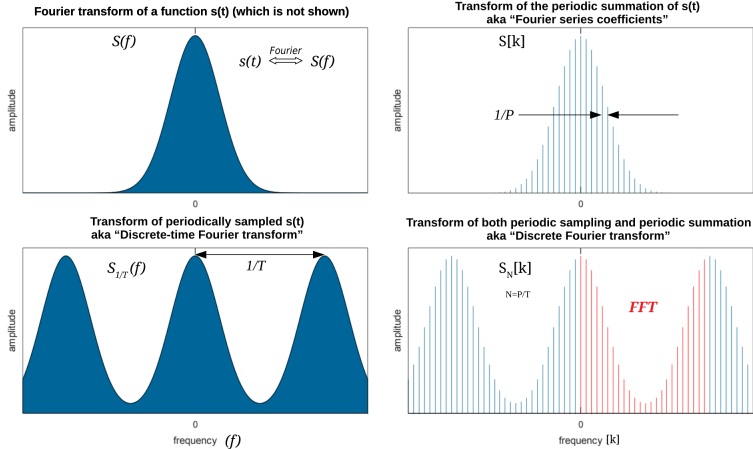
This process involves a fundamental trade-off: a longer window (N) provides better **frequency resolution** (ability to distinguish between close frequencies) but worse **time resolution** (ability to pinpoint when an event occurred).

Conversely, a shorter window improves time resolution at the expense of frequency resolution.

We use Fourier transforms to analyze the frequency content of signals. The resulting frequency information (spectra) is essential for:

- ▶ **Building Spectrograms:** Visual representations of how the frequency content of a signal changes over time.
- ▶ **Extracting Features:** Deriving key characteristics from a signal that can be used for tasks like speech recognition, music information retrieval, and audio classification.

Fourier Transforms You'll Actually Use (cont.)



Depiction of a Fourier transform. Source: Wikipedia

- ▶ STFT is a method to analyze the frequency content of a signal over time.
- ▶ It involves applying the Fourier Transform to short overlapping segments (frames) of the signal.
- ▶ The STFT provides a time-frequency representation of the signal.

► What:

- Slide a window across the signal.
- Compute a DFT per frame.

► Continuous form:

$$X(t, \omega) = \int x(\tau) w(\tau - t) e^{-j\omega\tau} d\tau$$

► Discrete idea:

- DFT of windowed frame at each time step.

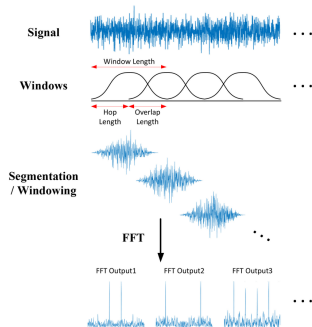
► How:

- Choose window: Hann / Hamming
- Frame length: e.g., 25 ms
- Hop: e.g., 10 ms

► Why:

- Speech is quasi-stationary in short frames.
- STFT captures how spectrum changes over time.

Short-Time Fourier Transform (STFT) (cont.)



STFT process: Signal divided into frames, each transformed into frequency domain.

Source: H. Jeon, Y. Jung, S. Lee and Y. Jung, "Area-Efficient Short-Time Fourier Transform Processor for Time-Frequency Analysis of Non-Stationary Signals," *Applied Sciences* 10(20):7208, 2020, Fig. 1 (CC BY 4.0).

- ▶ Written out, the STFT is a function of *two* variables: frame position n and frequency f ,

$$X(n, f) = \sum_{m=-\infty}^{\infty} x[m] w[m - n] e^{-j2\pi fm}$$

where $w[m - n]$ is the analysis window centred on frame n , and f is in cycles per sample.

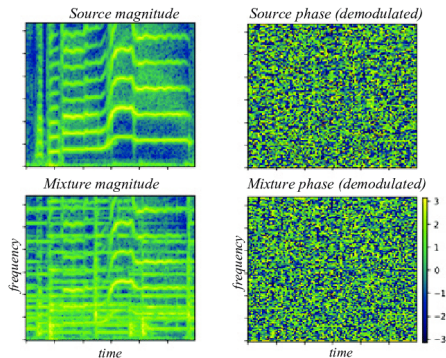
- ▶ What is actually computed is a framed DFT: with frame length N and hop H ,

$$X[n, k] = \sum_{m=0}^{N-1} x[m + nH] w[m] e^{-j2\pi km/N}, \quad k = 0, \dots, N-1$$

so the output is indexed by frame n and frequency bin k .

- ▶ The result is a complex-valued matrix, where each column represents the frequency content of a frame.
- ▶ The magnitude of the STFT gives the amplitude of each frequency at each time step.
- ▶ The phase of the STFT provides information about the timing of the frequency components.

Short-Time Fourier Transform (STFT) (cont.)



Magnitude vs. phase for a clean source (top) and a mixture (bottom). Harmonic and formant structure is visible in the magnitude; the demodulated phase looks close to noise, which is why most pipelines feed magnitude only.

Source: N. Takahashi, P. Agrawal, N. Goswami and Y. Mitsufuji, "PhaseNet: Discretized Phase Modeling with Deep Neural Networks for Audio Source Separation," Interspeech 2018, Fig. 1.

- ▶ A spectrogram is a visual representation of the STFT.
- ▶ It displays time on the x-axis, frequency on the y-axis, and color intensity represents amplitude.
- ▶ Spectrograms are widely used in audio analysis, speech recognition, and music processing.

► What:

- Heatmap of energy vs. time (x-axis) and frequency (y-axis).
- Color represents magnitude (often in dB).

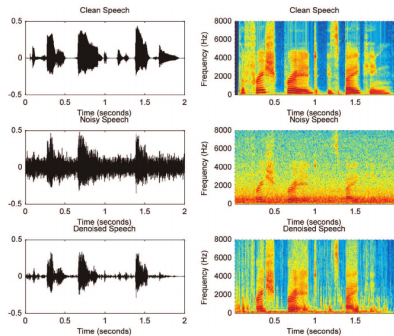
► How:

- Magnitude of STFT per time–frequency bin.
- Often log-scaled:

$$20 \log_{10} |X|$$

► Why:

- Visual patterns (formant tracks, harmonics, fricatives) → model-friendly features.
- Example: vowel formants appear as bright horizontal bands.

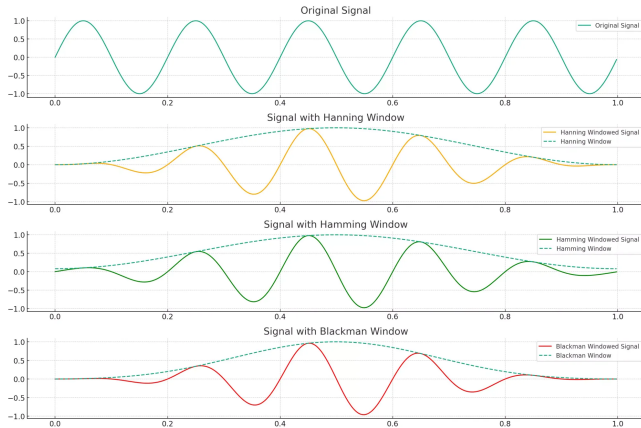


Example of a spectrogram: waveform (left) and time–frequency representation (right) of the same utterance, clean, noisy and denoised.

Source: H. Ye, G. Deng, S. J. Mauger, A. A. Hersbach, P. W. Dawson and J. M. Heasman, “A Wavelet-Based Noise Reduction Algorithm and Its Clinical Evaluation in Cochlear Implants,” *PLOS ONE* 8(9):e75662, 2013, Fig. 7 (CC BY).

- ▶ Spectrograms can be computed using various window functions (e.g., Hamming, Hanning) and different frame sizes.
- ▶ They provide insights into how the frequency content of a signal changes over time.

Spectrograms (cont.)



The same signal under a Hanning, Hamming, and Blackman window (dashed): each tapers the frame edges to zero, trading main-lobe width against side-lobe leakage.

Mel-Scale & Filterbanks

► What:

- Human pitch perception is **non-linear**.
- We perceive changes at low frequencies more finely than at high frequencies.
- Example: Difference between 200 Hz and 300 Hz feels larger than between 3000 Hz and 3100 Hz.

► How:

- Map raw frequency in Hz onto perceptual scales:
 - **Mel scale** – approximates human pitch perception.
 - **Bark scale** – based on critical bands of hearing.
 - **ERB scale** – Equivalent Rectangular Bandwidth, linked to auditory filter widths.
- Each scale compresses high-frequency regions while preserving low-frequency resolution.

► Why:

- Features reflecting human hearing are more effective for **Automatic Speech Recognition (ASR)** and **Text-to-Speech (TTS)**.
- Speech features (e.g., MFCCs) rely heavily on perceptual scales.

► Focus: Mel Scale

- Most widely used in speech/audio tasks.
- Roughly linear spacing below 1 kHz, logarithmic above.

► What:

- Mel scale converts frequency f (Hz) into perceptual pitch units (mels).
- Forward transform:

$$m = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

- Inverse transform:

$$f = 700 \left(10^{\frac{m}{2595}} - 1 \right)$$

- The constant 2595 is not arbitrary: it is chosen so that 1000 Hz lands on 1000 mels.

► How:

- Use Mel spacing to set filterbank center frequencies.
- Commonly used in **Mel Filterbanks** and **MFCCs**.

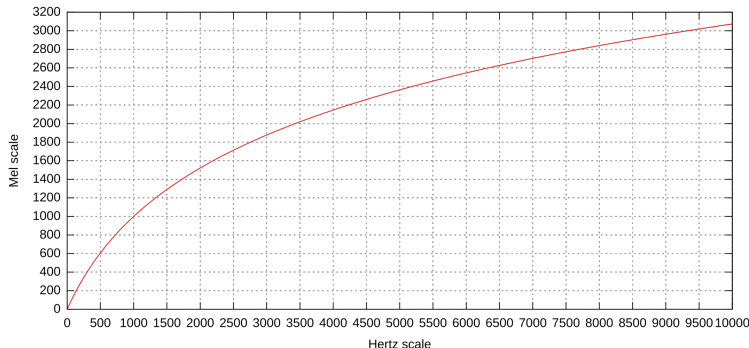
► Why:

- Aligns frequency resolution with **human pitch perception**.
- Compresses high-frequency regions (less perceptually important).
- Reduces unhelpful detail while retaining speech-relevant cues.

► Example:

- By convention, $1000 \text{ Hz} \approx 1000 \text{ mels}$.
- Below 1 kHz: roughly linear relation.
- Above 1 kHz: increasingly logarithmic.

Mel Scale: Definition & Intuition (cont.)

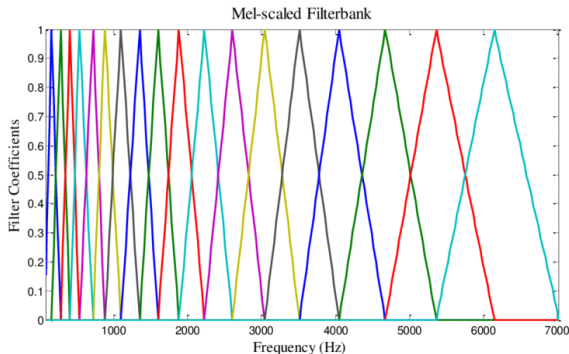


Plots of pitch mel scale versus hertz scale.

Source: Wikipedia, "Mel scale" (Wikimedia Commons, File:Mel-Hz plot.svg, by Krishna Vedala, CC BY-SA 3.0).

Mel Filterbanks:

- ▶ The frequency axis is warped to the Mel scale.
- ▶ Typically, N triangular filters (e.g., $N = 40$) are created, spaced evenly in the Mel domain.
- ▶ Each filter collects energy from a range of frequencies, emphasizing perceptually important bands.
- ▶ The output is a set of Mel filterbank energies, which are used as features for audio and speech processing tasks.



Mel filterbank basis functions, here using 20 Mel-filters over 0–7000 Hz.

Source: Yusnita M. A., Paulraj M. P., S. Yaacob, R. Yusuf and Shahrman A. B., "Analysis of Accent-Sensitive Words in Multi-Resolution Mel-Frequency Cepstral Coefficients for Classification of Accents in Malaysian English," *International Journal of Automotive and Mechanical Engineering*, vol. 7, pp. 1053–1073, 2013, Fig. 3.

Filterbank Construction Steps:

1. Convert the lower and upper frequency bounds to Mel scale.
2. Linearly space $N + 2$ points between these bounds in Mel scale.
3. Convert these points back to Hz to get filter edges.
4. For each filter k , define a triangular response:

$$H_k(f) = \begin{cases} 0, & f < f_{k-1} \\ \frac{f - f_{k-1}}{f_k - f_{k-1}}, & f_{k-1} \leq f < f_k \\ \frac{f_{k+1} - f}{f_{k+1} - f_k}, & f_k \leq f < f_{k+1} \\ 0, & f \geq f_{k+1} \end{cases} \quad (1)$$

5. Multiply the power spectrum by each filter and sum to get filterbank energies.

- ▶ Different feature types capture sound information at different levels:
 - **Spectrogram (linear-Hz):** closest to the raw physics of sound; preserves fine high-frequency detail.
 - **Mel-spectrogram:** compresses the high-frequency range using perceptual spacing, better reflecting human hearing.
 - **Log-Mel:** adds logarithmic compression to mimic loudness perception and control dynamic range.
 - **MFCCs (Mel-Frequency Cepstral Coefficients):** apply a DCT to Log-Mel features, creating compact, decorrelated coefficients widely used in classic ASR.

► Common processing pipeline:

Waveform $\rightarrow$ STFT $\rightarrow$ Mel scaling (optional) $\rightarrow$ log $\rightarrow$ DCT (optional)

► Why this matters:

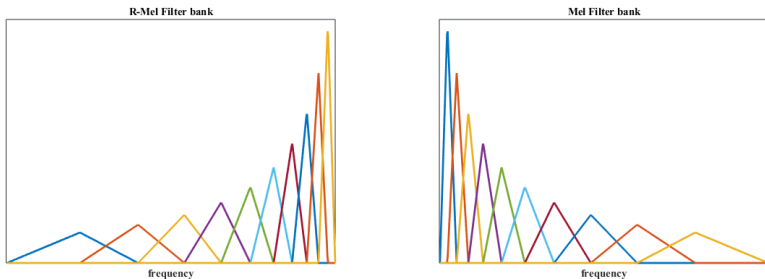
- More compact and stable features are easier for machine learning models to work with.
- Traditional models (like GMM-HMMs, SVMs) rely heavily on these, but even neural networks benefit when input dimensionality is reduced.

- ▶ A mel-spectrogram is a **time** $\times$ **mel-band matrix** created by re-binning STFT magnitudes through triangular filters on the Mel scale.
- ▶ The process:
 - Start with STFT $\rightarrow$ compute the power spectrum.
 - Apply a mel filterbank (typically 40–80 bands).
 - Take logarithms of the magnitudes to compress the dynamic range.
- ▶ Intuition:
 - Compact representation, aligned with human pitch perception.
 - Retains enough structure to capture speech/music content.
 - Works especially well with **CNNs** and **TTS acoustic models**.

- Equation (conceptual form):

$$E_m = \sum_k |X[k]|^2 H_m[k]$$

where H_m is the m -th triangular mel filter applied to the power spectrum.



Reversed-mel (left) and mel (right) filter banks.

Source: Wikimedia Commons, File:Mel RMel.png, by Ghasemzadehh, CC BY-SA 4.0.

- ▶ Log-Mel features are created by taking the logarithm of mel-band energies:

$$\log(E_m)$$

where E_m is the energy in the m -th mel band.

- ▶ Processing pipeline:

- Compute STFT $\rightarrow$ apply mel filterbank $\rightarrow$ mel-band energies.
- Apply logarithmic compression, either:

$$\log(E_m) \quad \text{or} \quad \log(1 + \alpha E_m)$$

to avoid issues with very small values.

► Why this step matters:

- Human perception of loudness is approximately logarithmic.
- Log scaling stabilizes feature magnitudes, preventing loud frames from dominating.
- Improves optimization for downstream machine learning models.

► Applications:

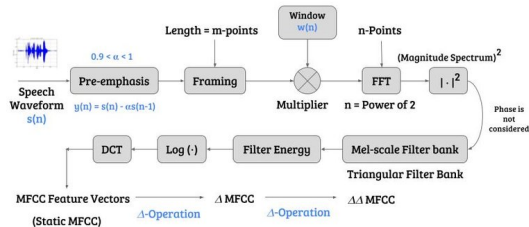
- Ubiquitous in modern **TTS systems**: log-mel spectrograms are the interface between the acoustic model and the vocoder (e.g., Tacotron 2 predicts log-mel frames, WaveNet turns them into a waveform).
- Widely used in speech recognition pipelines: Whisper takes 80-channel log-mel input.

- ▶ Mel-Frequency Cepstral Coefficients (MFCCs) provide a compact summary of the log-Mel spectrum, obtained by decorrelating features with a Discrete Cosine Transform (DCT).
- ▶ This representation captures the **spectral envelope** (linked to vocal-tract shape) while discarding fine pitch details.
- ▶ Typical processing pipeline:
 - (Optional) pre-emphasis to boost high frequencies.
 - Frame the signal, apply a window, and compute the STFT.
 - Apply a mel filterbank to obtain band energies.
 - Take logarithms of the energies.
 - Apply DCT to decorrelate; keep the first Q coefficients (e.g., 13).
 - Optionally append temporal derivatives (Δ) to capture dynamics.

► Why MFCCs are important:

- Historically very strong for HMM-GMM based ASR systems.
- Emphasize formant structure and vocal-tract characteristics.
- Tend to suppress pitch information, which is often less useful for recognition tasks.

- ▶ MFCCs are widely used in speech and audio processing.
- ▶ They capture the timbral texture of audio signals.
- ▶ Typically, only the first 13 coefficients are used for feature extraction.



A block diagram for MFCC computation: From audio signal to MFCC features.

Source: B. Barai, D. Das, N. Das, S. Basu and M. Nasipuri, "VQ/GMM-Based Speaker Identification with Emphasis on Language Dependency," *Advances in Intelligent Systems and Computing* 810, Springer, 2019, Fig. 2.

Pipeline:

1. Compute log-Mel spectrogram
2. Apply Discrete Cosine Transform (DCT)
3. Keep first 13 coefficients (MFCCs)

MFCC Equation:

$$c_k = \sum_{n=1}^N \log(E_n) \cos \left[\frac{\pi k}{N} \left(n - \frac{1}{2} \right) \right] \quad (2)$$

where:

- ▶ c_k is the k -th MFCC coefficient
- ▶ E_n is the energy in the n -th Mel filter
- ▶ N is the total number of Mel filters

- ▶ MFCCs are built from three main mathematical steps:

1. **Band energies (Mel filtering):**

$$E_m = \sum_k |X[k]|^2 H_m[k]$$

where H_m is the m -th triangular Mel filter.

2. **Logarithmic compression:**

$$L_m = \log(E_m + \epsilon)$$

which stabilizes values and mimics loudness perception.

3. **DCT-II (decorrelation):**

$$c_q = \sum_{m=1}^M L_m \cos \left[\frac{\pi q}{M} \left(m - \frac{1}{2} \right) \right], \quad q = 0, \dots, Q - 1$$

where M is the number of Mel bands and Q the number of coefficients (often $Q = 13$). The c_0 coefficient is sometimes dropped if energy is handled separately.

► Practical notes:

- Libraries like `librosa` or `torchaudio` implement this pipeline directly.
- Typical settings: $M = 40-80$ Mel bands, $Q \approx 13$ coefficients.

► Why this works:

- The DCT decorrelates the log-Mel energies.
- Produces compact, roughly uncorrelated features suited for traditional models.
- Historically central to HMM-GMM ASR pipelines.

Linear Predictive Coding (LPC):

- ▶ Models the speech signal as a linear combination of past samples.
- ▶ Useful for speech compression and speaker recognition.
- ▶ Captures the spectral envelope efficiently.

Perceptual Linear Prediction (PLP):

- ▶ Incorporates psychoacoustic models to mimic human hearing.
- ▶ Reduces spectral information to perceptually relevant features.
- ▶ Often used in robust speech recognition systems.

Pitch:

- ▶ Represents the fundamental frequency of speech.
- ▶ Important for prosody analysis, emotion detection, and speaker identification.

- ▶ Extracted using autocorrelation or cepstral methods.

Formants:

- ▶ Resonant frequencies of the vocal tract.
- ▶ Key for phoneme classification and speech synthesis.
- ▶ Estimated using LPC or spectral analysis.

Spectrogram-based Learned Features:

- ▶ Deep learning models (CNNs, RNNs) extract features directly from spectrograms.
- ▶ Capture complex patterns and temporal dependencies.
- ▶ Widely used in modern end-to-end speech and audio systems.

When to Choose What:

- ▶ **LPC/PLP:** When computational efficiency and interpretability are important; suitable for traditional speech tasks.
- ▶ **Pitch/Formants:** For tasks involving prosody, emotion, or phoneme-level analysis.
- ▶ **Spectrogram-based Learned Features:** For large-scale, data-driven applications where a deep model has enough data to learn its own filters from the spectrogram.
- ▶ Consider the task requirements, available data, and computational resources when selecting features.

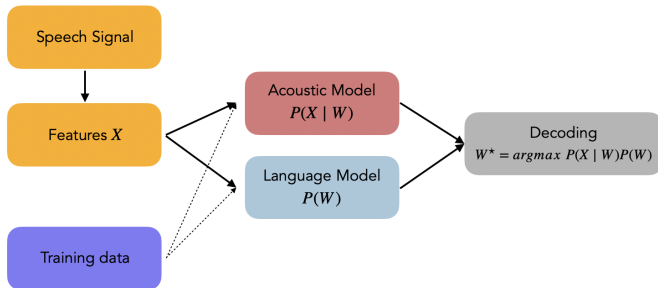
Automatic Speech Recognition (ASR)

- ▶ Automatic Speech Recognition converts spoken language (audio) into written text.
- ▶ Example:

Audio: “Hey computer, what’s the weather?” → Text: “hey computer what’s the weather”

- ▶ How it works (high-level pipeline):
 - Extract features from audio (e.g., log-Mel spectrograms, MFCCs).
 - Pass features through an **acoustic model** to map audio patterns to phonemes or subword units.
 - Use a **language model** to predict the most probable word sequences.
 - Decode the combined outputs into final text.

- ▶ Why ASR is important:
 - Enables natural language interaction with computers (voice assistants).
 - Facilitates transcription, accessibility, and spoken-input NLP applications.



Statistical historical approach to ASR.

Source: M. Fabien, "Introduction to Automatic Speech Recognition (ASR),"

https://maelfabien.github.io/machinelearning/speech_reco/

- ▶ Traditional ASR systems were built as modular pipelines with interpretable components.
- ▶ Main components:
 - **Acoustic model:** estimates $P(x \mid \text{phones})$, mapping audio features to phonetic units.
 - **Pronunciation dictionary:** converts phones to words.
 - **Language model:** estimates $P(\text{words})$, guiding likely word sequences.
 - **Decoder:** searches for the most probable word sequence given the acoustic and language models.
- ▶ Mathematical formulation:

$$\hat{W} = \arg \max_W P(W \mid X) = \arg \max_W P(X \mid W) P(W)$$

where W = word sequence, X = extracted audio features.

- ▶ Why this approach worked:

- Easier to engineer and understand.
- Interpretable: each module can be improved independently.

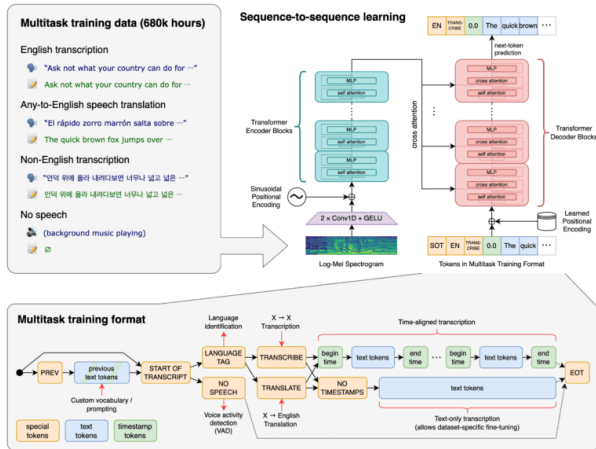
► Limitations:

- Complex, hand-crafted system.
- Errors can propagate through the pipeline.

- ▶ Modern ASR systems often replace traditional modular pipelines with **neural sequence-to-sequence models**, directly mapping audio features to text.
- ▶ Popular approaches:
 - **CTC (Connectionist Temporal Classification)** – handles variable-length alignments.
 - **Attention-based encoder-decoder** – learns soft alignments between input frames and output tokens.
 - **Transducer models** – integrate CTC-style alignment with recurrent decoding.
- ▶ Advantages:
 - Minimal handcrafting of features or components.
 - Often achieves better accuracy when trained on large datasets.
- ▶ Each approach solves the same problem, that the audio has far more frames than the transcript has tokens, in a different way. We take them one at a time in the sections that follow.

- ▶ Reference: Hannun, “Sequence Modeling With CTC”, Distill, 2017.
<https://distill.pub/2017/ctc/>

- ▶ Whisper is a state-of-the-art, multilingual ASR model developed by OpenAI.
- ▶ Key architecture:
 - Encoder-decoder transformer.
 - Trained on 680,000 hours of **weakly supervised** audio: transcripts scraped from the web and paired with audio, not annotated for the task. 117,000 of those hours are non-English.
 - Takes 80-channel log-Mel spectrograms as input (25 ms window, 10 ms hop).
- ▶ Advantages:
 - Robust to different accents and noisy environments.
 - Supports multitask learning: ASR and optional translation.
- ▶ Example use case:
 - Spoken French: “Je parle français” → can output English translation.



Source: Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision", 2022, Fig. 1.

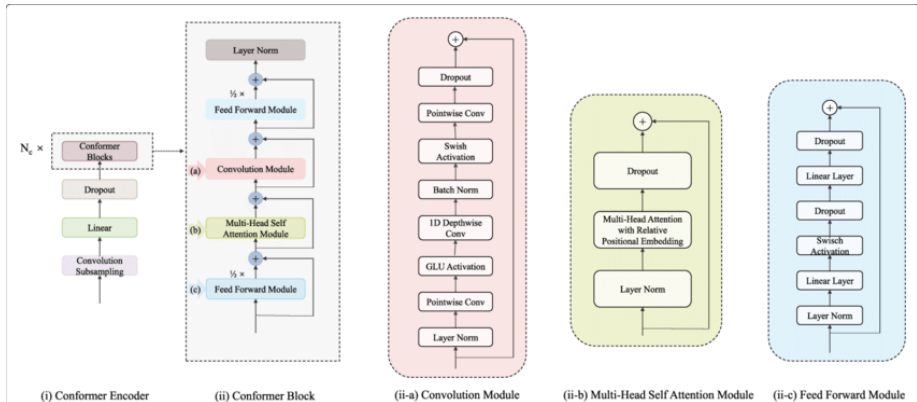
- ▶ Conformer is a hybrid neural model that combines **convolutions** and **transformers** to process speech effectively.
- ▶ Core components:
 - **Convolution:** captures local structures and fine-grained temporal patterns in audio.
 - **Self-attention:** captures global context and long-range dependencies.
- ▶ Achievements:
 - Was state of the art on LibriSpeech when published: 2.1% / 4.3% WER on test-clean / test-other without a language model, and 1.9% / 3.9% with an external LM (Conformer (L), 118.8M parameters).
 - Balances efficiency and accuracy better than pure transformer or CNN models alone: a 10.3M-parameter Conformer (S) still reaches 2.7% / 6.3%.

- Block equations: the four modules are applied *in sequence*, each as a pre-norm residual, with the two feed-forward modules half-weighted and sandwiching the attention and convolution pair:

$$\begin{aligned}\tilde{x} &= x + \frac{1}{2}\text{FFN}(x) & x' &= \tilde{x} + \text{MHSA}(\tilde{x}) \\ x'' &= x' + \text{Conv}(x') & y &= \text{LayerNorm}(x'' + \frac{1}{2}\text{FFN}(x''))\end{aligned}$$

where MHSA = multi-head self-attention (with relative positional encoding), Conv = convolutional module, FFN = feed-forward network.

Conformer (Conv + Transformer) (cont.)



Conformer encoder and its sub-modules, after Gulati et al., "Conformer", Interspeech 2020, Fig. 1. Redrawn version from Li et al., "Adversarial joint training with self-attention mechanism for robust end-to-end speech recognition", EURASIP J. Audio Speech Music Process., 2021.

- ▶ Whisper and Conformer represent two modern approaches to ASR with complementary strengths:
- ▶ **Whisper:**
 - Multilingual, trained on a massive weakly supervised dataset.
 - Robust to accents, noise, and varied conditions.
 - Open-sourced, general-purpose.
- ▶ **Conformer:**
 - Efficient hybrid model combining convolution + transformer.
 - Excels on English benchmarks like LibriSpeech.
 - Often fine-tuned for specific tasks.

► Trade-offs:

- Whisper is more general-purpose, handles diverse languages and noisy conditions.
- Conformer achieves very low WER on read English speech (2.1% on LibriSpeech test-clean without an LM) but may require task-specific tuning.

- ▶ Understanding the types of errors in ASR helps diagnose model weaknesses and improve performance.
- ▶ Common error types:
 - **Substitution:** a word is replaced by another. Example: “cat” → “cap”
 - **Deletion:** a word is dropped. Example: “the cat sat” → “cat sat”
 - **Insertion:** an extra word is added. Example: “cat sat” → “the cat sat”

- ▶ Several factors influence how well an ASR system performs in real-world conditions:
- ▶ **Noise level and microphone quality:** Background noise or poor recording devices can degrade feature extraction and recognition accuracy.
- ▶ **Speaker variability:** Differences in accent, pitch, speaking rate, or pronunciation can challenge the acoustic model.
- ▶ **Domain mismatch:** Models trained on casual speech may struggle with specialized domains (e.g., medical or legal speech).
- ▶ **Training data size and diversity:** More diverse and larger datasets generally improve robustness across speakers, accents, and environments.

CTC: Solving the Alignment Problem in Sequence-to-Sequence Tasks

Many sequence tasks (e.g., speech-to-text) face the **alignment problem**: input and output sequences are of different lengths and not aligned. CTC enables training without explicit alignment.

- ▶ **Input:** Sequence $\mathbf{x} = (x_1, x_2, \dots, x_T)$
- ▶ **Output:** Sequence $\mathbf{y} = (y_1, y_2, \dots, y_U)$
- ▶ **Challenge:** $T \neq U$, unknown alignment

CTC introduces a special blank token ($\emptyset$) and allows repeated labels. The final output is obtained by collapsing repeats and removing blanks. Note what the blank buys us: without it, a genuine double letter such as the “ll” in *hello* would collapse to a single “l”. A blank between the two repeats keeps them apart.

CTC: Training Without Alignments (cont.)



CTC: Blank token insertion and collapsing (the blank is written ϵ here).

Source: A. Hannun, "Sequence Modeling with CTC," *Distill*, 2017. <https://distill.pub/2017/ctc>

CTC Loss Function:

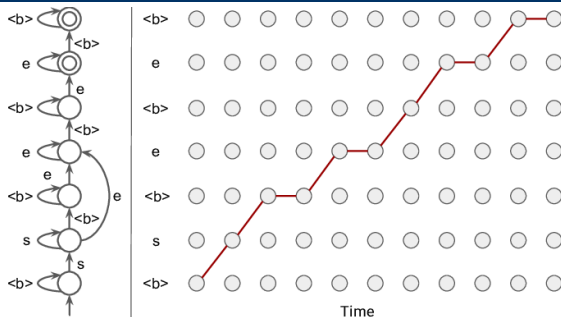
$$L_{CTC} = -\ln \left(\sum_{\pi \in \mathcal{B}^{-1}(\mathbf{y})} \prod_{t=1}^T p(\pi_t | \mathbf{x}) \right) \quad (3)$$

- ▶ π : a possible alignment (path) with blanks and repeats
- ▶ $\mathcal{B}^{-1}(\mathbf{y})$: set of all paths that collapse to $\mathbf{y}$
- ▶ $p(\pi_t | \mathbf{x})$: probability of label π_t at time t

CTC Decoding:

1. Predict a sequence of labels (including blanks) for each time step.
2. Collapse repeated labels and remove blanks to get the final output.

CTC: Training Without Alignments (cont.)



CTC alignment for the target "see": the automaton of all valid paths (left) and one alignment through the lattice (right); the blank is written .

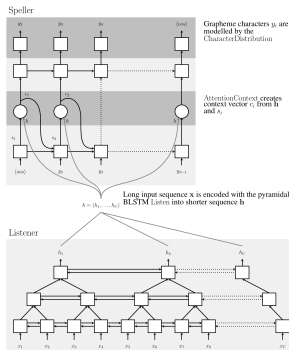
Source: R. Prabhavalkar, T. Hori, T. N. Sainath, R. Schlüter and S. Watanabe, "End-to-End Speech Recognition: A Survey," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 32, pp. 325–351, 2024, Fig. 1.

Encoder–Decoder Overview:

- ▶ **Encoder:** Processes the input acoustic features (e.g., Mel-spectrograms) and encodes them into a sequence of hidden representations.
- ▶ **Decoder:** Generates the output transcription, one token at a time, conditioned on the encoder outputs and previous decoder states.
- ▶ **Attention Mechanism:** Allows the decoder to focus on relevant parts of the input sequence at each decoding step, improving alignment and performance.
- ▶ Unlike CTC, nothing forces the alignment to be monotonic and no independence assumption is made across output tokens: the decoder is free to attend anywhere.

Listen, Attend and Spell (LAS): the canonical instance of this design.

- ▶ The **listener** is a pyramidal BLSTM encoder that shortens the frame sequence $\mathbf{x}$ (length T) into $\mathbf{h}$ (length $U \leq T$).
- ▶ The **speller** is an attention-based decoder emitting characters y_i one at a time.



Source: Chan et al., "Listen, Attend and Spell", ICASSP 2016, Fig. 1.

Attention Alignment:

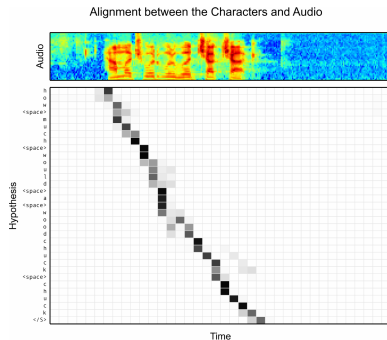
At each decoder time step t , the attention mechanism computes a context vector as a weighted sum of encoder outputs. The weights $\alpha_{t,s}$ represent the alignment between decoder step t and encoder position s :

$$\alpha_{t,s} = \frac{\exp(e_{t,s})}{\sum_{s'} \exp(e_{t,s'})} \quad (4)$$

where $e_{t,s}$ is the attention score (e.g., computed via a feedforward network) between decoder state at time t and encoder output at position s .

What the alignment looks like in practice

- ▶ Plotting $\alpha_{t,s}$ for every output character gives a heatmap of decoder step against audio frame.
- ▶ For speech it comes out close to diagonal, because speech is produced in order, even though nothing in the model enforces that.
- ▶ Where the model is unsure the mass spreads out: below, “woodchuck” draws a diluted band because “woodchuck” and “chuck” sound alike.

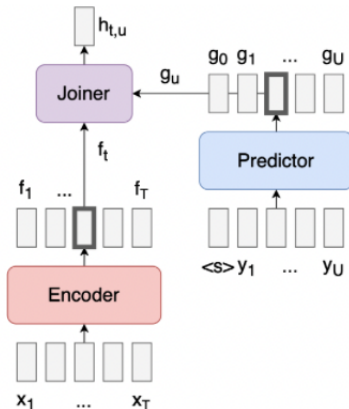


Alignment for the utterance “how much would a woodchuck chuck”. Source: Chan et al., “Listen, Attend and Spell”, ICASSP 2016, Fig. 2.

RNN-Transducer (RNN-T) is a sequence-to-sequence model widely used for end-to-end speech recognition. It extends the CTC (Connectionist Temporal Classification) approach by modeling both input and output dependencies:

- ▶ **Sequence-to-sequence model:** Maps input sequences (e.g., acoustic features) directly to output sequences (e.g., transcriptions).
- ▶ **End-to-end speech recognition:** Eliminates the need for separate components (like acoustic, pronunciation, and language models).
- ▶ **Extension of CTC:**
 - CTC models only input dependencies and assumes output tokens are conditionally independent.
 - RNN-T models both input and output dependencies, allowing the prediction of each output token to depend on previous outputs.
- ▶ **Key advantage:** Enables more accurate modeling of language and flexible alignments between input and output.

RNN-Transducer (RNN-T): Overview (cont.)



RNN-T architecture. The prediction network and joint network are also called the *predictor* and the *joiner*. Source: L. Lugosch, "Sequence-to-sequence learning with Transducers," 2020. <https://lorenlugosch.github.io/posts/2020/11/transducer/>

Architecture Components:

- ▶ **Encoder:** Processes the input acoustic features and produces high-level representations.
- ▶ **Prediction Network:** Acts like a language model, conditioning on previous non-blank output tokens.
- ▶ **Joint Network:** Combines encoder and prediction network outputs to produce logits over the output vocabulary (including the blank symbol).

$$\mathbf{h}_t = \text{Encoder}(\mathbf{x}_{1:t}) \quad (5)$$

$$\mathbf{g}_u = \text{PredictionNetwork}(y_{1:u-1}) \quad (6)$$

$$\mathbf{z}_{t,u} = \text{JointNetwork}(\mathbf{h}_t, \mathbf{g}_u) \quad (7)$$

Loss Overview:

- ▶ RNN-T loss marginalizes over all possible alignments between input and output sequences.
- ▶ Similar to CTC, but allows for output dependencies.
- ▶ Computed efficiently using dynamic programming.

$$\mathcal{L}_{\text{RNN-T}} = -\log \sum_{\pi \in \mathcal{A}(\mathbf{y}, T)} P(\pi | \mathbf{x}) \quad (8)$$

where $\mathcal{A}(\mathbf{y}, T)$ is the set of all valid alignments.

Pros vs. CTC:

- ▶ Models output dependencies (better language modeling).
- ▶ No need for external language model during inference.
- ▶ More flexible alignments.

Cons vs. CTC:

- ▶ More complex and computationally intensive: the loss is computed over a $T \times U$ lattice, so training memory grows with the product of input and output lengths.
- ▶ Harder to train and tune.
- ▶ Decoding is slower due to output dependencies.

Transformer & Conformer in ASR

- ▶ **Transformers** and **Conformers** are state-of-the-art architectures for Automatic Speech Recognition (ASR).
- ▶ Both use self-attention to model long-range dependencies in audio sequences.
- ▶ Conformer extends Transformer by integrating convolutional modules for local feature extraction.
- ▶ Key components include:
 - Multi-head self-attention
 - Feed-forward networks
 - Residual connections and layer normalization

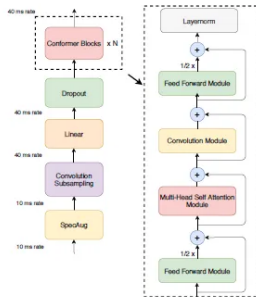


Figure 1: **Conformer encoder model architecture.** Conformer comprises of two macaron-like feed-forward layers with half-step residual connections sandwiching the multi-headed self-attention and convolution modules. This is followed by a post layernorm.

Source: Gulati et al., "Conformer", Interspeech 2020, Fig. 1.

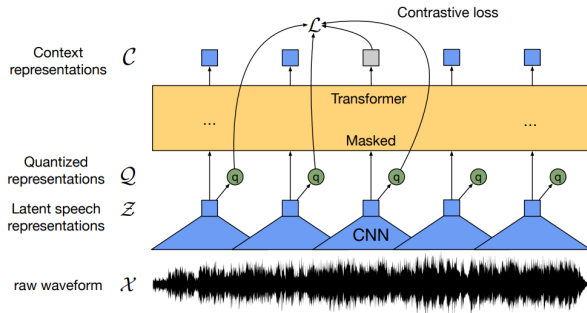
The encoder blocks are the ones covered earlier. Two things about speech force adjustments:

- ▶ **The sequence is very long.** At a 10 ms hop, one second of audio is 100 frames and a 30-second utterance is 3000, against perhaps 60 word-pieces of transcript. Self-attention costs $O(T^2)$, so ASR encoders first apply **convolutional subsampling**, typically two strided convolutions that cut the frame rate by $4\times$ (10 ms $\rightarrow$ 40 ms) before any attention layer runs.
- ▶ **Position matters relatively, not absolutely.** An utterance can start at any offset and last any length, so Conformer replaces absolute positional encoding with the **relative** sinusoidal scheme, making a block's behaviour depend on the distance between frames rather than their index.

Data augmentation on the input is also standard: **SpecAugment** masks bands of frequency and spans of time directly in the log-mel spectrogram.

- ▶ **Conformer** combines self-attention and convolution for effective speech modeling.
- ▶ **Block Structure:** two *half-step* feed-forward modules sandwich the attention and convolution pair (the “macaron” arrangement):
 - $\frac{1}{2}$ Feed-Forward $\rightarrow$ Multi-Head Self-Attention $\rightarrow$ Convolution $\rightarrow$ $\frac{1}{2}$ Feed-Forward $\rightarrow$ LayerNorm
- ▶ **Key Features:**
 - **Convolution Module:** LayerNorm $\rightarrow$ pointwise conv (expansion 2) $\rightarrow$ GLU $\rightarrow$ 1-D depthwise conv $\rightarrow$ BatchNorm $\rightarrow$ Swish $\rightarrow$ pointwise conv $\rightarrow$ dropout.
 - Every sub-module is a pre-norm residual unit; attention uses relative sinusoidal positional encoding.

- ▶ Proposed by Baevski et al. (2020): self-supervised learning for speech representation, the speech counterpart of BERT-style pretraining.
- ▶ **Architecture:** a CNN encoder turns the raw waveform into latent representations z_t ; a transformer produces context representations c_t ; a quantization module gives discrete targets q_t .
- ▶ **Pretraining:** mask spans of the latent sequence and train contrastively to pick the true quantized target among distractors, on unlabeled audio alone.



Source: Baevski et al., "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations", NeurIPS 2020, Fig. 1.

Fine-tuning for ASR with Wav2Vec 2.0

- ▶ **Self-Supervised Learning (SSL):** Pre-train Wav2Vec 2.0 on large unlabeled audio.
- ▶ **Fine-tuning:** Use small labeled dataset for ASR.
- ▶ **Loss Function:** Connectionist Temporal Classification (CTC) Loss.
- ▶ **Goal:** Map audio features to text transcriptions.

- ▶ **Small labeled data:** Only 10 minutes of transcribed speech.
- ▶ **Performance:** Word Error Rate (WER) 4.8% (test-clean) / 8.2% (test-other).
- ▶ **Read the small print.** That headline number needs all three of:
 - the `LARGE` model (~ 317 M parameters),
 - pre-training on 53k hours of unlabeled LibriVox audio (LV-60k),
 - a Transformer language model at decoding time.

Swap the LM for a 4-gram and the same model gives 6.6% / 10.3%; decode with no LM at all and it is around 40%. The `BASE` model at 10 minutes gives 6.9% / 12.9%.

- ▶ **Efficient:** High accuracy with minimal *supervision*, not with minimal compute or data.

Model	Unlabeled data	LM	dev		test	
			clean	other	clean	other
10 min labeled						
Discrete BERT [4]	LS-960	4-gram	15.7	24.1	16.3	25.2
BASE	LS-960	4-gram	8.9	15.7	9.1	15.6
		Transf.	6.6	13.2	6.9	12.9
LARGE	LS-960	Transf.	6.6	10.6	6.8	10.8
	LV-60k	Transf.	4.6	7.9	4.8	8.2
1h labeled						
Discrete BERT [4]	LS-960	4-gram	8.5	16.4	9.0	17.6
BASE	LS-960	4-gram	5.0	10.8	5.5	11.3
		Transf.	3.8	9.0	4.0	9.3
LARGE	LS-960	Transf.	3.8	7.1	3.9	7.6
	LV-60k	Transf.	2.9	5.4	2.9	5.8
10h labeled						
Discrete BERT [4]	LS-960	4-gram	5.3	13.2	5.9	14.1
Iter. pseudo-labeling [58]	LS-960	4-gram+Transf.	23.51	25.48	24.37	26.02
	LV-60k	4-gram+Transf.	17.00	19.34	18.03	19.92
BASE	LS-960	4-gram	3.8	9.1	4.3	9.5
		Transf.	2.9	7.4	3.2	7.8
LARGE	LS-960	Transf.	2.9	5.7	3.2	6.1
	LV-60k	Transf.	2.4	4.8	2.6	4.9

Source: Baevski et al., “wav2vec 2.0”, NeurIPS 2020, Table 1 (extract).

Wav2Vec 2.0 + CTC Loss (PyTorch)

```
import torch
import torchaudio
from transformers import Wav2Vec2ForCTC, Wav2Vec2Processor

processor = Wav2Vec2Processor.from_pretrained("facebook/wav2vec2-base-960h")
model = Wav2Vec2ForCTC.from_pretrained("facebook/wav2vec2-base-960h")

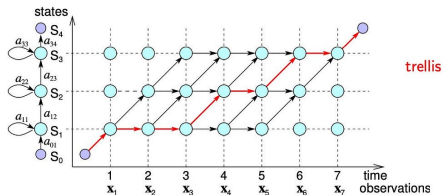
input_audio, _ = torchaudio.load("audio.wav")
inputs = processor(input_audio.squeeze(), sampling_rate=16000,
                   return_tensors="pt")
with torch.no_grad():
    logits = model(**inputs).logits
pred_ids = torch.argmax(logits, dim=-1)
transcription = processor.decode(pred_ids[0])
print(transcription)
```

► CTC Decoding:

- Greedy decoding: Select most probable token at each timestep.
- Beam search: Explore multiple hypotheses for better accuracy.
- Viterbi algorithm: Find most likely sequence.

► Equation:

$$\hat{y} = \arg \max_y P(y|x)$$



$$p(\mathbf{X}, \text{path}_\ell | \lambda) = p(\mathbf{X} | \text{path}_\ell, \lambda) P(\text{path}_\ell | \lambda)$$

$$\text{likelihood: } \sum_{\{\text{path}_\ell\}} p(\mathbf{X}, \text{path}_\ell | \lambda)$$

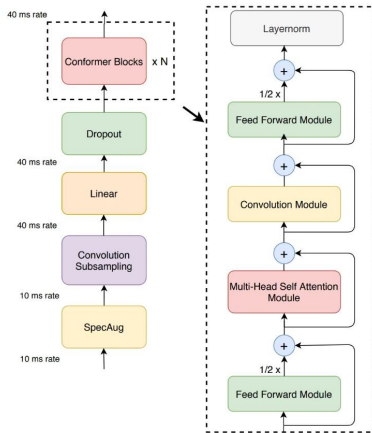
$$\text{decode: } \max_{\text{path}_\ell} p(\mathbf{X}, \text{path}_\ell | \lambda)$$

The same sum-over-paths (likelihood) vs. max-over-paths (Viterbi) choice, drawn on an HMM trellis.

Source: H. Shimodaira and S. Renals, "Hidden Markov Models and Gaussian Mixture Models," Automatic Speech Recognition Lectures 4–5, University of Edinburgh.

Conformer-CTC: Non-autoregressive ASR

- **Conformer:** Combines convolutional and transformer layers.
- **CTC Loss:** Connectionist Temporal Classification for sequence alignment.
- **Non-autoregressive:** Processes entire input sequence in parallel.



Source: Gulati et al., "Conformer," Interspeech 2020,

Fig. 1.

► CTC (Connectionist Temporal Classification):

- Aligns input and output sequences without explicit segmentation.
- Every output token is predicted independently, so all frames decode in parallel.
- Simpler and faster inference; benefits most from an external language model.

► Transducer Models:

- Jointly model alignment and output prediction.
- Autoregressive in the output: each token conditions on the previous ones.
- Better for streaming and online decoding, since the encoder need not see the whole utterance.
- More complex, higher accuracy in some cases.

- ▶ **NVIDIA Riva / NeMo:** production ASR toolkits shipping pre-trained Conformer-CTC checkpoints, optimized for GPU inference and both real-time and batch processing.
- ▶ **Parakeet:** a family of NeMo models built on a **FastConformer** encoder, a redesigned Conformer with a more aggressive downsampling schema that runs about $2.8\times$ faster, paired with a CTC, RNN-T, or TDT (token-and-duration transducer) decoder.
- ▶ **Citrinet:** a fully convolutional non-autoregressive CTC model, 1-D time-channel separable convolutions plus squeeze-and-excitation, from 9.8M to 142M parameters. Fast because CTC decoding is non-autoregressive; Riva now marks it superseded by Parakeet and Conformer-CTC.
- ▶ **References:**
 - [Conformer \(arXiv:2005.08100\)](#), [FastConformer \(arXiv:2305.05084\)](#), [Citrinet \(arXiv:2104.01721\)](#)
 - [NVIDIA Riva ASR model docs](#)

- ▶ **SpeechBrain:** open-source toolkit for speech processing.
- ▶ Supports streaming ASR through a Conformer *transducer* trained with dynamic chunk training, so the encoder only ever sees a bounded chunk of future audio. CTC is kept as an auxiliary loss to stabilise training.
- ▶ Released streaming checkpoints for LibriSpeech and GigaSpeech.



The three output shapes SpeechBrain recipes cover: a class label, a continuous value, and a transcript. Source: SpeechBrain (Ravanelli et al., 2021).

- ▶ Conformer-CTC enables fast, parallel ASR.
- ▶ CTC models are simpler and efficient, at the cost of assuming output tokens are independent.
- ▶ NVIDIA NeMo/Riva and SpeechBrain provide production implementations.
- ▶ Streaming needs bounded lookahead, which is why streaming recipes usually reach for a transducer rather than plain CTC.

Spoken Language Understanding (SLU)

- ▶ SLU focuses on understanding the meaning of spoken utterances, going beyond mere transcription.
- ▶ Goal: transform “what was said” → “what was meant.”
- ▶ Input and output:
 - Input: audio signal.
 - Output: structured meaning, typically **intent** + **slots**.
- ▶ Definitions:
 - **Intent**: the action the user wants to perform (e.g., play music, set an alarm).
 - **Slots**: arguments of the intent (e.g., {song: “Yesterday”, artist: “Beatles”}).

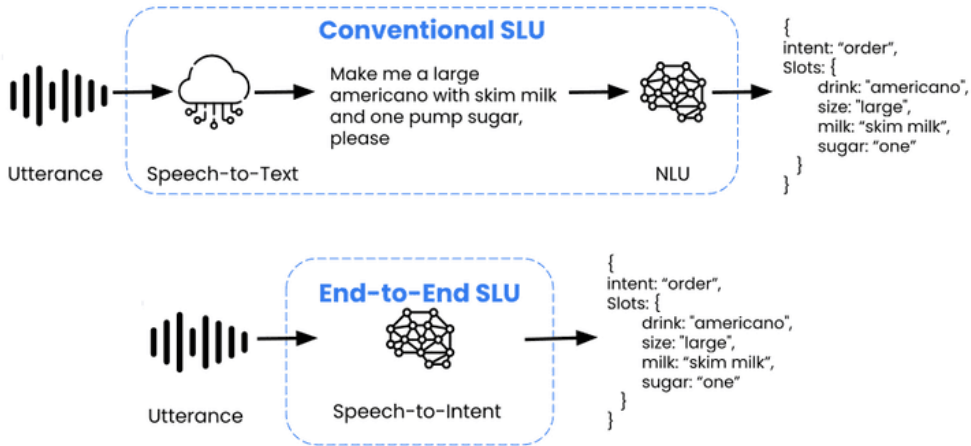
► Example:

- Audio: "Set an alarm for 7 a.m."
- Output: {intent: set_alarm, time: 7:00}

► Why SLU is important:

- Enables interaction with smart assistants and dialogue systems.
- Bridges the gap between raw speech signals and actionable NLP tasks.

Spoken Language Understanding (SLU) (cont.)



Spoken Language Understanding. Source: [PICOVOICE.AI](https://picovoice.ai)

- ▶ Many commercial SLU systems follow a two-step pipeline, separating speech recognition from understanding.
- ▶ Pipeline steps:
 1. **Audio → ASR → Text Transcript** Converts raw audio into written words.
 2. **Text → NLU Model**
 - ▶ **Intent classification:** identifies user's action (e.g., `request_weather`).
 - ▶ **Slot filling:** extracts entities or arguments (e.g., {location: "London"}).
 3. **Dialogue manager:** decides system response or action based on intent and slots.

► Advantages of this approach:

- Reuses existing NLP models trained on text.
- Easier to build and maintain using large text corpora.

► Example:

- User says: "What's the weather like in Paris?"
- ASR output: "what's the weather like in paris"
- NLU output: {intent: get_weather, location: "Paris" }

- ▶ Direct SLU models bypass transcription and predict meaning directly from audio signals.
- ▶ How it works:
 - Input features: log-Mel spectrograms or wav2vec embeddings.
 - Model architecture: typically CNNs or Transformers for classification or sequence labeling.
 - Output: intent + slots, directly mapped from audio.

► Advantages:

- Avoids error propagation from ASR mistakes.
- Ideal for domain-specific commands where transcription is unnecessary.
- Faster and more efficient than pipeline approaches.

► Example:

- Audio: “Play jazz”
- Direct SLU output: {intent: play_music, genre: “jazz” }
- No intermediate transcript is generated.

- ▶ Spoken Language Understanding faces several real-world challenges:
- ▶ **Error Propagation:** In pipeline systems, ASR mistakes can cause NLU errors. *Example:* “Play rock” misheard as “Play doc” → system fails.
- ▶ **Data Scarcity:** Annotated SLU datasets are often small. Collecting paired (speech, intent, slots) data is expensive.
- ▶ **Multilinguality & Code-Switching:** Users often mix languages in the same utterance. *Example:* “Ponme una canción by Shakira.” Systems must handle hybrid language input robustly.
- ▶ **Context & Dialogue:** Understanding multi-turn conversations requires memory of previous interactions. *Example:* “Play jazz.” → “Now make it louder.”

- ▶ Spoken Language Understanding powers many real-world applications by extracting meaning from speech, not just transcribing it.
- ▶ Common use cases:
 - **Virtual assistants:** Siri, Alexa, Google Assistant.
 - **Smart homes:** voice-controlled lights, appliances.
 - **Customer service:** routing calls based on intent.
 - **Automotive:** navigation, calls, climate control.
 - **Healthcare:** converting doctors' spoken notes into structured forms.

► Example (Alexa):

- User says: “Turn off the bedroom lights.”
- SLU output: {intent: control_device, device: lights, location: bedroom}

► Why SLU is critical:

- Transcription alone isn't sufficient; systems must act on the meaning of what was said.

Advanced Tasks: Diarization & Emotion Recognition

► **Goal:** Identify “who spoke when” in audio streams

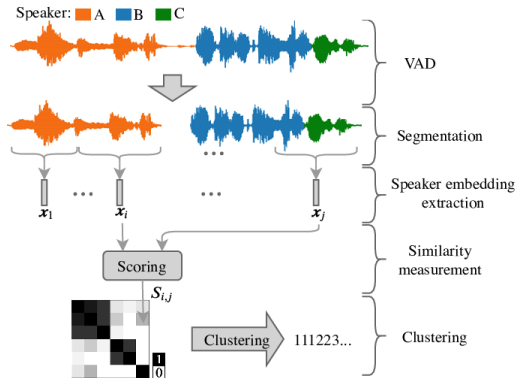
► **Applications:**

- Meeting transcription
- Podcast analysis
- Call center monitoring

► **Pipeline:**

1. **Voice activity detection (VAD):** separate speech from silence and noise
2. **Segmentation:** cut the speech into windows, or detect speaker change points
3. **Embedding:** map each segment to a speaker vector (x-vector, ECAPA-TDNN)
4. **Clustering:** group segments by speaker from pairwise similarities
5. **Resegmentation:** refine the boundaries and assign overlapping speech

VAD comes first for a reason: any speech it drops is lost for good, and any noise it admits gets assigned to some speaker. Its errors propagate through every later stage.

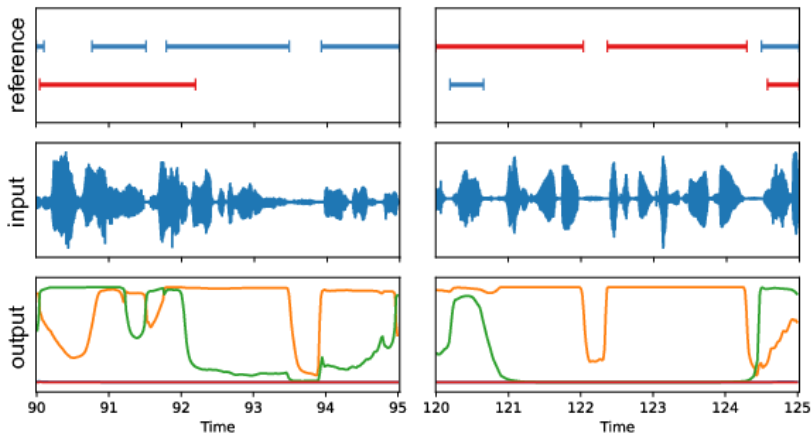


Source: Q. Lin, Y. Hou and M. Li, "Self-Attentive Similarity Measurement Strategies in Speaker Diarization," Interspeech 2020,

Fig. 1.

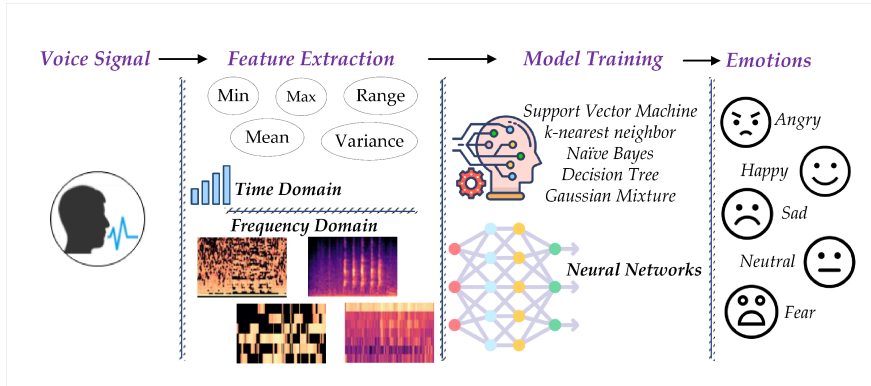
- ▶ **Popular Tool:** `pyannote.audio` (Bredin et al., ICASSP 2020)
- ▶ **Features:**
 - Pre-trained diarization models
 - Easy integration with Python
 - Supports custom pipelines
- ▶ Recent versions replace the modular chain above with an end-to-end neural segmentation model, so the classical pipeline is now one option rather than the only one.

Speaker Diarization Tools (cont.)



Reference annotation (top) against the model's frame-wise speaker posteriors (bottom). Source: `pyannote.audio`.

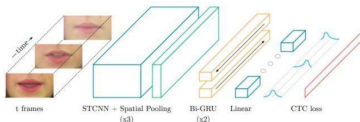
- ▶ **Goal:** Detect emotions from speech signals
- ▶ **Applications:**
 - Human-computer interaction
 - Mental health monitoring
 - Customer service analytics
- ▶ **Datasets:** IEMOCAP, RAVDESS, MSP-IMPROV, CREMA-D



The classical SER pipeline. Source: A. S. Alluhaidan, O. Saidani, R. Jahangir, M. A. Nauman and O. S. Neffati, "Speech Emotion Recognition through Hybrid Features and Convolutional Neural Network," *Applied Sciences* 13(8):4750, 2023, Fig. 1 (CC BY 4.0).

- ▶ The standard recipe is a **self-supervised speech encoder plus a light classifier head**:
 - **Wav2Vec 2.0, HuBERT, WavLM**: general-purpose encoders, robust to noise and speaker variation
 - **emotion2vec**: pre-trained on emotion data specifically
 - Frames are pooled (mean or attention pooling) and passed to a small classifier
- ▶ **Performance on IEMOCAP** (4-class: angry, happy, neutral, sad):
 - ~63–68% accuracy with the encoder *frozen* and only a linear head trained
 - ~73–75% once the encoder itself is *fine-tuned*
- ▶ **Compare numbers with care.** IEMOCAP results are almost always reported on the 4-class collapse with “excited” merged into “happy”, under leave-one-session-out splits, and papers differ on whether they quote weighted (WA) or unweighted (UA) accuracy. Two figures from different papers are usually not comparable.

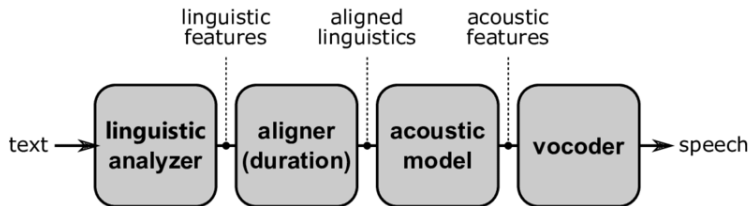
- ▶ **LipNet** (Assael et al., 2016) decodes sentences from lip movement alone, with no audio at all.
- ▶ **Architecture:** spatiotemporal convolutions over the video frames $\rightarrow$ bidirectional GRU $\rightarrow$ CTC loss, trained end to end. The same CTC machinery as in audio ASR, with video frames in place of spectrogram frames.
- ▶ **Result:** 95.2% sentence-level accuracy on the GRID corpus (overlapped-speaker split), against 86.4% for the previous word-level state of the art.
- ▶ **Why it matters:** it gives a channel that survives when the audio does not. Fusing the two, *audio-visual* speech recognition, improves accuracy in noise.



Source: Assael et al., "LipNet: End-to-End Sentence-level Lipreading", 2016, Fig. 1.

- ▶ Speech synthesis (Text-to-Speech, TTS) converts written text into spoken audio: the inverse problem of ASR.
- ▶ **Pipeline:** text analysis $\rightarrow$ acoustic model (text to mel-spectrogram) $\rightarrow$ vocoder (mel-spectrogram to waveform). Almost every named system below is one of those two stages, not a whole system.
- ▶ **Three eras:**
 - concatenative / unit-selection and statistical-parametric (HMM-based) synthesis;
 - autoregressive neural: Tacotron 2 (acoustic model) driving a WaveNet vocoder;
 - parallel and diffusion neural: FastSpeech 2 (non-autoregressive acoustic model) or Grad-TTS (diffusion acoustic model), paired with a parallel vocoder such as HiFi-GAN.
- ▶ These pair up rather than compete: Grad-TTS in fact uses HiFi-GAN as its vocoder. Fully end-to-end systems such as VITS collapse both stages into one model.

- **Applications:** accessibility and screen readers, voice assistants, audiobooks, navigation, language learning.



Source: H.-T. Luong, "Deep learning based voice cloning framework for a unified system of text-to-speech and voice conversion," PhD thesis, SOKENDAI, 2020, Fig. 2.1.

- ▶ Modern TTS systems face several practical and ethical challenges:
- ▶ **Prosody & Emotion:** Generating natural intonation, emphasis, and expressive speech remains difficult.
- ▶ **Cross-Speaker Synthesis:** Voice cloning or multi-speaker synthesis requires adapting models to new voices without losing quality.
- ▶ **Low-Resource Languages:** Limited data makes it challenging to build high-quality TTS systems for many languages.
- ▶ **Ethical Issues:** Risk of misuse in deepfakes or generating fake celebrity voices.

Evaluation

- ▶ Word Error Rate (WER) is a standard metric to evaluate ASR transcription accuracy. It is based on the edit distance between the reference and the hypothesis.
- ▶ Formula:

$$\text{WER} = \frac{S + D + I}{N}$$

where:

- S = substitutions
- D = deletions
- I = insertions
- N = total words in reference

► Example:

- Reference: “the cat sat on the mat”
- Hypothesis: “the cat sat mat”
- Errors: 2 deletions $\rightarrow \text{WER} = 2/6 \approx 33.3\%$

► Why it matters:

- Provides an intuitive, widely-used measure of transcription accuracy for ASR systems.

► Two things to keep in mind:

- The denominator counts *reference* words, so insertions are unbounded and **WER can exceed 100%**. A model that hallucinates a paragraph over a one-word utterance scores far above 1.
- WER is computed after normalisation (case, punctuation, numbers), and different normalisers give different WERs on the same output. Comparisons across papers are only meaningful when the normaliser matches.

- ▶ **Sentence error rate:** the whole utterance is right or wrong. One substitution costs the entire sentence, so it is too strict to steer development — which is why WER's local credit won.
- ▶ **Phoneme and frame error rates:** the same edit-distance idea at finer granularity; a phoneme error rate needs a pronunciation dictionary, a frame error rate needs a time alignment.
- ▶ **Real-time factor (RTF):** processing time divided by audio duration — $RTF < 1$ means faster than real time. Together with latency, it decides whether a model can serve a live interface at all.
- ▶ **Downstream metrics:** when ASR feeds another system, score the pipeline end to end — e.g., BLEU after speech translation, task success for a voice assistant.
- ▶ **Tooling:** NIST's SCTK (`sc-lite`) has standardized WER scoring — alignment, normalization, significance tests — for decades; use it rather than reimplementing edit distance.

WER answers one question: did we get the words right. A diarization system with excellent WER can still be useless if it attributes every turn to the wrong speaker, and a synthesizer can be word-perfect and still sound wrong. Each task needs its own metric.

- **CER (Character Error Rate)** is the same edit distance counted over characters instead of words:

$$\text{CER} = \frac{S_c + D_c + I_c}{N_c}$$

Used where word boundaries are not marked in the writing system (Chinese, Japanese, Thai), and for character-level output models, where a one-character slip should not cost a whole word.

- **DER (Diarization Error Rate)** scores “who spoke when”:

$$\text{DER} = \frac{T_{\text{false alarm}} + T_{\text{missed}} + T_{\text{confusion}}}{T_{\text{total speech}}}$$

that is, non-speech scored as speech, plus speech scored as non-speech, plus speech given to the wrong speaker, over total reference speech time. Like WER it can exceed 100%. Always state whether a forgiveness *collar* (commonly 0.25 s) was applied around reference boundaries and whether overlapping speech was scored: DER is not comparable across those settings.

- ▶ **MOS (Mean Opinion Score)** is the reference metric for synthesis, and it is a *human* measurement. Listeners rate each utterance in isolation on a five-point scale, 5 excellent, 4 good, 3 fair, 2 poor, 1 bad, and the score is the mean over ratings (ITU-T Rec. P.800). TTS work adapts the scale to ask about **naturalness** rather than transmission quality.
 - MOS is not comparable across studies: report the number of listeners, the number of utterances, and confidence intervals.
 - Good systems now sit close to recorded speech, so plain MOS saturates. Side-by-side comparative MOS (CMOS) or AB preference tests separate them better.
 - Objective proxies exist but do not replace it: mel-cepstral distortion (MCD), which needs time-aligned ground-truth audio and so only works when reconstructing held-out recordings; ASR word error rate on synthesized speech as an intelligibility check; and learned MOS predictors.

Challenges & Current Research

- ▶ **Robustness:** Handling noise, accents, and low-quality microphones.
- ▶ **Compute Cost of SSL:** Pre-training wav2vec-style models consumes GPU-months of compute; cheaper self-supervised recipes remain open.
- ▶ **Interpretability & Data Bias:** Understanding model decisions and correcting demographic skews in training corpora.
- ▶ **Cross-Modal Fusion:** Combining speech with text and vision (audio-visual ASR, spoken multimodal assistants) is still brittle.
- ▶ **Multilinguality & Code-Switching:** Supporting diverse populations and hybrid language input.
- ▶ **Low-Resource Speech:** Many languages lack sufficient training data for ASR/TTS models.
- ▶ **Expressive Speech Synthesis:** Capturing prosody, emotion, and enabling style transfer.

- ▶ **Evaluation:** Moving beyond BLEU/WER to semantic, human-centered, and LLM-based metrics.
- ▶ **Ethics:** Addressing risks like deepfake voices and protecting privacy in speech datasets.

- ▶ Speech is a rich modality for natural language processing, bridging sound and meaning.
- ▶ Audio features such as **MFCC**, **Mel**, and **Log-Mel** form the foundation for ASR and TTS systems.
- ▶ The spoken interaction cycle involves **ASR** → **SLU** → **TTS**, enabling complete end-to-end communication.
- ▶ **Evaluation**: WER measures transcription accuracy for ASR; human listening tests remain the reference for synthesis quality.
- ▶ Future directions:
 - More **end-to-end models** for robust and expressive speech understanding/synthesis.
 - Improved, **semantic and human-centered evaluation metrics**.
 - Stronger **ethical safeguards** against misuse.

References

- [1] Jurafsky, D., & Martin, J. H. (2026). *Speech & Language Processing* (3rd edition draft).
<https://web.stanford.edu/~jurafsky/slp3/>
- [2] Rabiner, L., & Juang, B.-H. (1993). *Fundamentals of Speech Recognition*. Prentice Hall.
- [3] Deng, L., & Li, X. (2013). Machine learning paradigms for speech recognition: An overview. *IEEE Transactions on Audio, Speech, and Language Processing*, 21(5), 1060–1089.
- [4] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
- [5] Kubichek, R. F. (1993). Mel-cepstral distance measure for objective speech quality assessment. *IEEE Pacific Rim Conference on Communications, Computers and Signal Processing (PACRIM)*, 125–128.
- [6] Rabiner, L. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286.

- [7] Graves, A., Fernández, S., Gomez, F., & Schmidhuber, J. (2006). Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. *ICML*, 369–376.
- [8] Hannun, A. (2017). Sequence modeling with CTC. *Distill*. <https://distill.pub/2017/ctc/>
- [9] Graves, A. (2012). Sequence transduction with recurrent neural networks. *arXiv preprint arXiv:1211.3711*.
- [10] Chan, W., Jaitly, N., Le, Q. V., & Vinyals, O. (2016). Listen, attend and spell: A neural network for large vocabulary conversational speech recognition. *ICASSP*, 4960–4964. arXiv:1508.01211.
- [11] Park, D. S., Chan, W., Zhang, Y., Chiu, C.-C., Zoph, B., Cubuk, E. D., & Le, Q. V. (2019). SpecAugment: A simple data augmentation method for automatic speech recognition. *Interspeech*, 2613–2617. arXiv:1904.08779.
- [12] Gulati, A., Qin, J., Chiu, C.-C., et al. (2020). Conformer: Convolution-augmented transformer for speech recognition. *Interspeech*, 5036–5040. arXiv:2005.08100.

- [13] Rekish, D., Koluguri, N. R., Krizan, S., et al. (2023). Fast Conformer with linearly scalable attention for efficient speech recognition. *IEEE ASRU*, 1–8. arXiv:2305.05084.
- [14] Majumdar, S., Balam, J., Hrinchuk, O., Lavrukhin, V., Noroozi, V., & Ginsburg, B. (2021). Citrinet: Closing the gap between non-autoregressive and autoregressive end-to-end models for automatic speech recognition. *arXiv preprint arXiv:2104.01721*.
- [15] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*.
- [16] Baevski, A., Zhou, H., Mohamed, A., & Auli, M. (2020). wav2vec 2.0: A framework for self-supervised learning of speech representations. *NeurIPS 33*. arXiv:2006.11477.
- [17] Hsu, W.-N., Bolte, B., Tsai, Y.-H. H., Lakhota, K., Salakhutdinov, R., & Mohamed, A. (2021). HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29, 3451–3460. arXiv:2106.07447.

- [18] Chen, S., Wang, C., Chen, Z., et al. (2022). WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6), 1505–1518. arXiv:2110.13900.
- [19] Wang, Y., et al. (2017). Tacotron: Towards end-to-end speech synthesis. *arXiv preprint arXiv:1703.10135*.
- [20] Shen, J., Pang, R., Weiss, R. J., et al. (2018). Natural TTS synthesis by conditioning WaveNet on mel spectrogram predictions. *ICASSP*, 4779–4783. arXiv:1712.05884.
- [21] van den Oord, A., et al. (2016). WaveNet: A generative model for raw audio. *arXiv preprint arXiv:1609.03499*.
- [22] Ren, Y., Hu, C., Tan, X., Qin, T., Zhao, S., Zhao, Z., & Liu, T.-Y. (2021). FastSpeech 2: Fast and high-quality end-to-end text to speech. *ICLR*. arXiv:2006.04558.
- [23] Kong, J., Kim, J., & Bae, J. (2020). HiFi-GAN: Generative adversarial networks for efficient and high fidelity speech synthesis. *NeurIPS 33*. arXiv:2010.05646.
- [24] Popov, V., Vovk, I., Gogoryan, V., Sadekova, T., & Kudinov, M. (2021). Grad-TTS: A diffusion probabilistic model for text-to-speech. *ICML*, PMLR 139, 8599–8608. arXiv:2105.06337.

- [25] Kim, J., Kong, J., & Son, J. (2021). VITS: Conditional variational autoencoder with adversarial learning for end-to-end text-to-speech. *ICML*. arXiv:2106.06103.
- [26] Park, T. J., Kanda, N., Dimitriadis, D., Han, K. J., Watanabe, S., & Narayanan, S. (2022). A review of speaker diarization: Recent advances with deep learning. *Computer Speech & Language*, 72, 101317. arXiv:2101.09624.
- [27] Lin, Q., Hou, Y., & Li, M. (2020). Self-attentive similarity measurement strategies in speaker diarization. *Interspeech*, 284–288.
- [28] Bredin, H., Yin, R., Coria, J. M., et al. (2020). pyannote.audio: Neural building blocks for speaker diarization. *ICASSP*, 7124–7128.
- [29] Busso, C., Bulut, M., Lee, C.-C., et al. (2008). IEMOCAP: Interactive emotional dyadic motion capture database. *Language Resources and Evaluation*, 42(4), 335–359.
- [30] Chen, L.-W., & Rudnicky, A. (2023). Exploring wav2vec 2.0 fine tuning for improved speech emotion recognition. *ICASSP*, 1–5. arXiv:2110.06309.

- [31] Ma, Z., Zheng, Z., Ye, J., Li, J., Gao, Z., Zhang, S., & Chen, X. (2024). emotion2vec: Self-supervised pre-training for speech emotion representation. *Findings of ACL 2024*, 15747–15760. arXiv:2312.15185.
- [32] Assael, Y. M., Shillingford, B., Whiteson, S., & de Freitas, N. (2016). LipNet: End-to-end sentence-level lipreading. *arXiv preprint arXiv:1611.01599*.
- [33] Chung, J. S., Senior, A., Vinyals, O., & Zisserman, A. (2017). Lip reading sentences in the wild. *CVPR*. arXiv:1611.05358.
- [34] Ravanelli, M., et al. (2021). SpeechBrain: A general-purpose speech toolkit. *arXiv preprint arXiv:2106.04624*.
- [35] NVIDIA Riva, ASR models reference. <https://docs.nvidia.com/deeplearning/riva/user-guide/docs/reference/models/asr.html>
- [36] ITU-T Recommendation P.800 (1996). *Methods for Subjective Determination of Transmission Quality*. International Telecommunication Union, Geneva.
- [37] Papineni, K., Roukos, S., Ward, T., & Zhu, W.-J. (2002). BLEU: A method for automatic evaluation of machine translation. *ACL*.

- [38] Panayotov, V., Chen, G., Povey, D., & Khudanpur, S. (2015). LibriSpeech: An ASR corpus based on public domain audio books. *ICASSP*.
- [39] Ardila, R., et al. (2020). Common Voice: A massively-multilingual speech corpus. *LREC*, 4218–4222. arXiv:1912.06670.
- [40] Hernandez, F., Nguyen, V., Ghannay, S., Tomashenko, N., & Estève, Y. (2018). TED-LIUM 3: Twice as much data and corpus repartition for experiments on speaker adaptation. *Speech and Computer (SPECOM)*, LNCS 11096, 198–208. Springer.
- [41] Carletta, J., et al. (2005). The AMI meeting corpus: A pre-announcement. *MLMI*, LNCS 3869, 28–39. Springer.
- [42] LJSpeech dataset. <https://keithito.com/LJ-Speech-Dataset/>
- [43] CSTR VCTK Corpus (version 0.92). <https://datashare.ed.ac.uk/handle/10283/3443>
- [44] Bakhturina, E., Lavrukhin, V., Ginsburg, B., & Zhang, Y. (2021). Hi-Fi multi-speaker English TTS dataset. *arXiv preprint arXiv:2104.01497*. Download: <https://www.openslr.org/109/>

- [45] ATIS intent/slot dataset. <https://www.kaggle.com/datasets/siddhadev/ms-cntk-atis>
- [46] SLURP dataset (audio hosted on Zenodo). <https://github.com/pswietojski/slurp>
- [47] Whisper ASR Demo. <https://huggingface.co/spaces/openai/whisper>
- [48] Tacotron 2 + WaveGlow Demo.
<https://colab.research.google.com/github/NVIDIA/DeepLearningExamples/blob/master/PyTorch/SpeechSynthesis/Tacotron2/notebooks/Tacotron2.ipynb>
- [49] Google Cloud Text-to-Speech. <https://cloud.google.com/text-to-speech>
- [50] Coqui TTS (maintained fork). <https://github.com/idiap/coqui-ai-TTS>
- [51] Li, J. (2022). Recent advances in end-to-end automatic speech recognition. *APSIPA Transactions on Signal and Information Processing*, 11(1), e8. arXiv:2111.01690.
- [52] Prabhavalkar, R., Hori, T., Sainath, T. N., Schlüter, R., & Watanabe, S. (2024). End-to-end speech recognition: A survey. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 32, 325–351. arXiv:2303.03329.

- [53] Qin, L., Xie, T., Che, W., & Liu, T. (2021). A survey on spoken language understanding: Recent advances and new frontiers. *IJCAI*, 4577–4584. [arXiv:2103.03095](#).
- [54] Tan, X., Qin, T., Soong, F., & Liu, T.-Y. (2021). A survey on neural speech synthesis. *arXiv preprint arXiv:2106.15561*.